Waiz Khan

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Education

Johns Hopkins University — Baltimore, MD
M.S.E. in Data Science — *Aug. 2024 – May 2027*
- Coursework: Applied Statistics and Data Analysis, Introduction to Data Science, Deep Learning, Artificial Intelligence, Investment Science, Financial Derivatives

GD Goenka University — Gurugram, India
B.Tech. in Computer Science & Engineering — *Aug. 2019 – Nov. 2023*
- Coursework: Data Mining, Distributed Systems, Linear Algebra, Stochastic Processes

Research Experience

Research Assistant — Dec. 2025 – Present
Johns Hopkins University, Advisor: Prof. Philipp Koehn — *Baltimore, MD*

Measuring the Language Share of Multilingual Representations — *First author, submitted to ICLR 2027*
- Built the analysis pipeline: internal representations from 17 open language models (0.6B to 9B parameters) across 128 languages, run as batch jobs on a Slurm cluster
- Designed Language Factor Share, a variance decomposition that measures how much of a model's internal representation reflects the language versus the meaning; all 17 models show the same dip in the middle layers, and the pattern replicates on a second dataset (FLORES-200, rank correlation 0.92)
- Found that the models' shared middle is not English: the average of the other languages predicts each language better than English does for 113 to 124 of 127 languages
- Confirmed 4 of 4 blind predictions on held out model families, and showed the statistic fails as a training target: pushing it down also hurt cross language alignment ($r = 0.92$ over 18 runs)

Ancient Script Decipherment
- Audited a 12 corpus ancient script dataset, found roughly 34% of the Egyptian evaluation data duplicated from training, and rebuilt it as 29 frozen splits; on the cleaned data, label blind analysis still recovered sign function above chance (ARI 0.169, $p = 0.0015$)

Experience

Co-Founder & CTO — Apr. 2026 – Present
Iris, healthcare AI | JHU Pava Center Accelerator, Spark Program | ARPA-H ADVOCATE 2026 proposal — *Baltimore, MD*
- Built the data layer (TypeScript, Node.js, PostgreSQL) that combines records from several clinical sources into one versioned patient record, with provenance on every value and a full change history
- Built automated checks that block any AI generated claim that cannot be traced back to source records, run as a merge gate so no change ships without passing them

Projects

OFX: Real Time Market Data Pipeline | *TypeScript, Node.js* — Jun. 2026 – Present
- Ingests live exchange feeds, rebuilds the full order book, detects gaps and resyncs automatically, and checks itself against independent exchange snapshots; stores clean, normalized events in daily partitions for replay
- Computes order flow features (microprice, queue and depth imbalance, order flow imbalance) and a study harness that tests whether they predict short term price moves, with a chronological split, scaling fit on training data only, and Newey–West standard errors

AttendOpt | *Python, scikit-learn, CatBoost, DoWhy, PuLP* — 2022, rebuilt 2026
- Began as research with Dr. Yogesh Kumar (GD Goenka University). Merged IHDS-II survey files into a 47,027 student dataset and compared 8 models on splits grouped by state, so no household appears in both train and test (best ROC-AUC 0.841 vs. 0.699 for logistic regression)
- Reported ranking and calibration separately: the best ranking model (CatBoost) was not the best calibrated (random forest, ECE 0.039), which matters when a score decides who gets help
- Tested the distance to school effect with DoWhy refutation checks (a placebo treatment drops it from 0.14 to 0.001) and allocated a fixed intervention budget with an integer program

Technical Skills

Languages: Python, SQL, TypeScript, C++, Bash
Data & ML: pandas, NumPy, SciPy, scikit-learn, XGBoost, CatBoost, PyTorch, Hugging Face Transformers, PostgreSQL
Methods: exploratory analysis, feature engineering, classification and regression, calibration, hypothesis testing (bootstrap, permutation), causal inference, time series validation, integer programming, Monte Carlo
Infrastructure: Linux, Slurm HPC, Git, Docker, REST APIs